

# Response to Reviewers

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*“Benchmarking Deep Reinforcement Learning for Cartesian Path Tracking of Free-Floating Space Robot Manipulators”*

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Dear Dr. Fornés and Reviewers,

We thank the Associate Editor and the three reviewers for their careful reading and constructive feedback. All comments have been addressed in the revised manuscript. Below, each comment is restated verbatim and followed by our response and the concrete action taken. In the revised manuscript, changes are highlighted per reviewer: Reviewer 1 changes in yellow, Reviewer 2 changes in red, Reviewer 3 changes in blue, and changes addressing reviewers 2 + 3 in green. Section and equation references below refer to the revised manuscript.

## Reviewer 1

We thank Reviewer 1 for the positive assessment and for considering the work a valuable contribution. The single clarification request is addressed below.

**R1.1 Comment:** *III.A: from the description I do not understand why AI is mentioned for the digital twin: it seems more a typical simulation scenario with injected noise and uncertainty. Please clarify.*

**Response:** We agree the original wording did not make the distinction clear. The “AI” aspect of the digital twin does not refer to injected noise or stochastic disturbances, but to the fact that the digital twin hosts a learning-based control agent (a deep reinforcement-learning PPO policy) that autonomously synthesizes control strategies from interaction, rather than executing a pre-specified controller. This is what distinguishes the digital twin from a conventional simulation scenario.

**Action taken:** Section III.A was rewritten to clarify this point. The revised text explains that the information model of the digital twins provides a structured, interoperable description (kinematics, inertial properties, joint configurations, constraints), while the AI capability is embodied by the DRL agent that is trained and evaluated through interaction with the twins. It now states explicitly that the twins serve as adaptive training and validation environments rather than reproducing predefined behavior, and that scenarios/constraints are sampled from bounded uniform distributions during training.

## Reviewer 2

**R2.1 Comment:** *Identify the gaps in the existing research on RL-based control of space robotics systems.*

**Response:** The research gaps are now stated explicitly at the end of the related-work discussion.

**Action taken:** Added a dedicated paragraph identifying three mutually reinforcing gaps: (i) no prior work systematically compares multiple RL algorithms under identical conditions for Cartesian end-effector tracking of free-floating space robots; (ii) base attitude minimization, a safety-critical requirement, is omitted in most RL-based contributions; and (iii) formal-methods-based safety monitoring is largely absent. This appears in Section II (Related Work).

**R2.2 Comment:** *Organize previous studies according to classical control techniques, reinforcement learning-based control techniques, hybrid safety-aware control systems, and benchmarking studies according to advantages and limitations.*

**Response:** The related-work section was restructured into the four requested categories, each discussing advantages and limitations.

**Action taken:** Section II now covers, in separate subsections: (1) classical model-based control (computed torque, resolved-motion rate, reactionless planning); (2) RL-based control (path planning, obstacle avoidance, trajectory tracking, failure management); (3) hybrid safety-aware control (shielding, control barrier functions); and (4) comparative/benchmarking studies. Advantages and limitations are discussed for each category.

**R2.3 Comment:** *More information can be added to the table of the related research, such as the robot's DOF, the simulator used, the structure of the reward functions, the structure of the observation/action space, the safety limits, and the metrics used for the evaluation.*

**Response:** Table 1 was expanded with the requested information. One item was deliberately not included as a column: the structure of the reward functions varies considerably across the surveyed works and is in most cases only partially reported, so a compact tabular representation would have been either misleading or unreadable. Reward design in the surveyed literature is instead discussed in the text of Section II.

**Action taken:** Table 1 now reports, for each surveyed work, the RL algorithm, degrees of freedom (DOF), simulator, observation-/action-space dimensions, safety limits, formal verification, and evaluation metrics, with an accompanying abbreviation legend.

**R2.4 Comment:** *Provide a definition of the reward function with specific weight values, followed by the explanation of the physical intuition behind each reward component.*

**Response:** The reward function is now given with concrete weight values and a per-component physical justification.

**Action taken:** The revised Section IV states all weights explicitly (e.g.  $W_p = 150$  for end-effector position error,  $W_v = 25$  for velocity error, etc.) and adds a paragraph explaining the physical intuition of each term — e.g. the position-error weight dominates because trajectory tracking is the primary objective, while the velocity weight is kept lower to avoid over-penalizing transient dynamics.

**R2.5 Comment:** *Carry out a sensitivity analysis on the reward weight values to verify the stability of the algorithm's ordering over various reward definitions.*

**Response:** A one-at-a-time sensitivity analysis of the dominant reward weights was added. (This comment overlaps with R3.4 and is addressed jointly.)

**Action taken:** New subsection ("Reward Weight Sensitivity Analysis") with an accompanying table: the four dominant weights were each scaled by  $\times 2$  and  $\times 0.5$  around their baseline, the PPO agent retrained, and KPIs recomputed over 30 evaluation episodes. The results show the agent still converges and the qualitative KPI conclusions are robust to moderate miscalibration.

**R2.6 Comment:** *Add an ablation study to investigate the impact of each reward shaping component on the final result.*

**Response:** An ablation study of the reward components was added.

**Action taken:** New subsection ("Ablation Study of Reward Components") with a dedicated table. Each of the seven weights in the cost term was set to zero in turn while the remaining six were held at their baseline values, and the optimized PPO agent was retrained for each variant. In addition, the two positive reward terms were disabled separately (progress signal  $r_{prog}$  and proximity bonus  $r_{bonus}$ ), yielding nine ablation variants in total. KPIs are averaged over 30 evaluation episodes on the circular trajectory, quantifying each component's contribution. The results identify  $W_{ori}$  and  $W_p$  as the two dominant cost components.

**R2.7 Comment:** *Tune the failure penalties based on the reward scale, followed by the explanation of the impact on policy learning.*

**Response:** The terminal failure penalty is now explicitly matched to the reward scale, with a discussion of its effect on learning.

**Action taken:** The revised text sets the terminal penalty  $r_{fail} = -1$  and explains it is matched to the normalized reward scale (normalization constant  $C = 500$ , so typical step rewards lie in a comparable range). The impact on policy learning — discouraging premature termination without dominating the gradient — is discussed.

**R2.8 Comment:** *Clearly describe the number of training runs, seed values, and the policy selection procedure.*

**Response:** The experimental protocol is now fully specified.

**Action taken:** Added: all runs use MATLAB's Mersenne Twister generator with fixed seed `rng(0,'twister')`; three independent training runs per algorithm (18 in total across six algorithms) under identical conditions; the policy checkpoint with the highest mean episodic reward over the final 50 training episodes is selected and used for all evaluations.

**R2.9 Comment:** *Apply comparable hyperparameter tuning procedures to all reinforcement learning algorithms rather than optimizing only PPO.*

**Response:** We agree that comparable tuning across all algorithms would strengthen the study, and we did attempt exactly that. A Bayesian hyperparameter optimization was set up in the Reinforcement Learning Designer to apply an identically budgeted search (tuning goal: average episodic reward) to all six agents. The workflow could not be completed. With parallel execution enabled, the tuning spawned workers beyond the physical core count and no training episode ever started; without parallel execution, MATLAB either terminated mid-training with an internal error originating in the Simulink engine, or deadlocked indefinitely when initializing the second optimization trial. The behaviour was reproduced on MATLAB R2025b and R2026a, and additionally on MATLAB Online to exclude a local installation issue. It was confirmed for TRPO and TD3; since a like-for-like comparison requires all six agents, the search was not pursued further. A support request documenting these failure modes has been filed with MathWorks.

We note that this affects only the automated tuning workflow. Standard training is unaffected: all 18 baseline training runs and the 16 tuning runs for PPO completed without incident. The PPO tuning reported in the manuscript was carried out as a manual coordinate-wise grid search, which is not practical to replicate across six algorithms: with sequential execution only, and per-run training times of 8 to 38 minutes per 1000 episodes, an equally budgeted manual search across all agents amounts to several days of uninterrupted wall-clock time.

We would also like to point out that the comparison as reported is not asymmetric. All six algorithms are run under identical, toolbox-recommended default hyperparameters, so no algorithm is granted a tuning budget the others do not receive. The reported ranking is a ranking under default settings, and the PPO tuning is presented as a separate follow-up study on the algorithm that performed best in that baseline, not as part of the six-way comparison itself. Reporting tuned results for a subset of algorithms would have introduced precisely the asymmetry this comment asks us to avoid.

**Action taken:** The wording of the comparison claim in the Conclusion was qualified to state that the six-algorithm comparison holds under unified conditions and identical default hyperparameters. No further changes were made: the comparison was already conducted under identical defaults, and the PPO tuning is already presented separately from it. Per-algorithm tuning under an identical search budget is left to future work.

**R2.10 Comment:** *Describe the hyperparameter optimization process for PPO including search space, optimization method, and tuning.*

**Response:** The PPO optimization process is now described in full. (Overlaps with R3.8, addressed jointly.)

**Action taken:** Added a “Hyperparameter Tuning” description: a coordinate-wise grid search in which each hyperparameter was swept across a discrete candidate set while the others were held at their current best, selecting the configuration with the highest mean episodic reward over the final 50 of 1000 training episodes (16 tuning runs total). The swept parameters, candidate values, selected values, and the resulting network architecture ([256,256,128] policy network) are tabulated.

**R2.11 Comment:** *Evaluate robustness by introducing disturbances, sensor noise, parameter uncertainties, and actuator saturation variations.*

**Response:** A dedicated robustness stress-test campaign was added.

**Action taken:** New subsection (“Robustness Evaluation under Non-Ideal Conditions”) with a dedicated table. The trained optimized PPO policy was evaluated without retraining over 30 episodes for each of five non-ideal conditions on the circular trajectory: sensor noise injected between the observation vector and the agent, reduced actuator authority (torque limits lowered to  $0.75, \tau_{max}$ ), an external joint-torque disturbance applied after the saturation block, parameter uncertainty, and a combined stress case applying several of these simultaneously. All KPIs are reported for each condition.

**R2.12 Comment:** *Try to add plots such as learning curves, error distributions, and comparative performance charts.*

**Response:** Learning curves, error distributions, and comparative charts were added.

**Action taken:** Added training/learning-curve figures, confidence-interval charts (95% bootstrap CIs for selected KPIs across all agents), and comparative performance plots.

**R2.13 Comment:** *Visualize results using boxplots, violin plots, and confidence intervals to better illustrate performance variability.*

**Response:** Boxplots and confidence intervals were added to illustrate variability.

**Action taken:** Added boxplots of selected KPI distributions over the 30 evaluation episodes per agent (IQR box, median line,  $1.5 \times \text{IQR}$  whiskers, outliers) alongside the 95% bootstrap confidence-interval figures, highlighting the spread differences between stable (PPO, TRPO) and unstable methods.

**R2.14 Comment:** *Revise the conclusion to align with the actual experimental evidence.*

**Response:** The conclusion was rewritten to reflect the actual results.

**Action taken:** The revised conclusion summarizes the six-algorithm comparison identifying PPO as the best trade-off, reports the concrete quantitative gains of the optimized PPO agent (including on the additional piecewise-linear reference: return K1 from -0.4565 to -0.2977, K4 base-orientation error reduced by 44%, energy K9 reduced by 35%), and states conclusions strictly supported by the evidence under the stated assumptions.

**R2.15 Comment:** *Improve grammar, sentence structure, and clarity through careful proofreading.*

**Response:** The full manuscript was carefully proofread. In contrast to the other revisions, these language edits are *not* colour-marked in the revised manuscript. The proofreading touched nearly every paragraph, and marking it would have obscured the content-related changes requested by the reviewers, which are the ones the highlighting is meant to make traceable. We would like to emphasise that these edits are purely linguistic: wording, sentence structure, and readability were improved, while the technical meaning of the sentences was preserved throughout.

**Action taken:** Grammar, sentence structure, and clarity were revised throughout the entire manuscript. These edits are left unmarked to keep the reviewer-specific highlighting legible; no statement, claim, or result was altered in the process.

**R2.16 Comment:** *Enhance the quality of all figures.*

**Response:** Figure quality was improved throughout.

**Action taken:** Figures were regenerated/enhanced for resolution, labeling, and legibility.

## Reviewer 3

**R3.1 Comment:** *The novelty compared to existing RL-based robotic control studies should be more explicitly highlighted; clearly differentiate the contributions from prior works listed in Table 1.*

**Response:** The novelty statement was sharpened and tied directly to Table 1.

**Action taken:** Added text following Table 1 explaining that surveyed works typically evaluate a single RL algorithm on a specific subtask, and that none jointly addresses the three compounding challenges tackled here: a multi-algorithm benchmark under identical conditions, base-attitude minimization, and formal safety monitoring. This differentiates our contribution from each prior entry in Table 1.

**R3.2 Comment:** *Assumptions such as ideal sensing, absence of noise, and perfect simulation fidelity need clearer justification; discuss how these assumptions affect real-world deployment.*

**Response:** All idealizing assumptions are now explicitly stated and justified, with deployment implications discussed.

**Action taken:** Added a paragraph enumerating the assumptions (ideal sensing with state read directly from Simulink signals, no sensor noise/latency, rigid-body model, no external disturbances) and justifying them as standard for fair RL comparison and reproducibility. Their effect on real deployment is discussed in the Limitations/Future Work subsection (see R3.9).

**R3.3 Comment:** *The paper would benefit from a comparison with other simulation platforms (e.g., Gazebo, MuJoCo).*

**Response:** A comparison of the chosen platform with common alternatives was added.

**Action taken:** Added a discussion contrasting MATLAB/Simulink/Simscape with Gazebo and MuJoCo, including a summary table. The comparison acknowledges the advantages of the alternatives — MuJoCo's highly optimized contact and multibody solver and its status as a de-facto RL benchmark standard, Gazebo's mature ROS integration and plug-in-based sensor models — and states the decisive reason for the present choice: neither offers a native path to the formal-verification ecosystem this work relies on. Simscape Multibody provides a validated solver for the free-floating base–manipulator coupling and integrates directly with Stateflow and the Simulink Design Verifier used to specify the runtime safety monitors, yielding a single

reproducible toolchain from physical model through learned policy to formally checked safety logic. The trade-off — higher raw throughput in MuJoCo, richer sensor and ROS tooling in Gazebo — is stated explicitly.

**R3.4 Comment:** *The selection of weight parameters appears empirical; a sensitivity analysis or justification for these weights would improve reproducibility and rigor.*

**Response:** Addressed jointly with R2.5 via the new reward-weight sensitivity analysis and the per-weight physical justification (R2.4).

**Action taken:** See the new sensitivity-analysis subsection (weights scaled  $\times 2$  and  $\times 0.5$ , PPO retrained, KPIs recomputed) and the per-component physical intuition added to the reward definition.

**R3.5 Comment:** *The scalability of the safety mechanisms to real-time onboard systems is not discussed; include computational overhead analysis.*

**Response:** A computational-overhead and scalability discussion was added.

**Action taken:** New paragraph (“Computational Overhead and Scalability”): the three safety monitors (momentum observer, collision-distance checker, joint-limit assertion) consist of simple algebraic operations over link states scaling as  $O(n_{\text{links}})$  per step, negligible relative to the physics simulation, supporting real-time onboard feasibility.

**R3.6 Comment:** *Some metrics such as energy consumption and smoothness could be better justified in the context of space missions.*

**Response:** The operational significance of these KPIs for space missions is now justified.

**Action taken:** Added justification: control smoothness (K7) captures torque chattering that accelerates actuator wear, excites structural vibrations degrading pointing accuracy of cameras/star trackers, and wastes energy; energy consumption (K9) is justified by the strict power budgets of on-orbit operations.

**R3.7 Comment:** *Statistical significance tests (e.g., confidence intervals or hypothesis testing) are missing; including these would validate whether observed performance differences are meaningful.*

**Response:** Non-parametric significance testing was added.

**Action taken:** New subsection (“Statistical Significance of Agent Comparisons”). A Friedman test over the 30 evaluation episodes per agent confirms that at least one agent differs significantly for every KPI (all  $p < 10^{-17}$ ), justifying pairwise comparisons. Pairwise Wilcoxon signed-rank tests between PPO (baseline) and each competing agent follow, with Holm correction applied to control the family-wise error rate across multiple comparisons. Holm-corrected p-values and median differences are tabulated, together with 95% bootstrap confidence intervals, establishing which performance differences are statistically meaningful.

**R3.8 Comment:** *The tuning process appears heuristic; a systematic hyperparameter search strategy (e.g., grid search, Bayesian optimization) should be discussed for reproducibility.*

**Response:** Addressed jointly with R2.10: the systematic search procedure is now described.

**Action taken:** See the coordinate-wise grid-search description (search space, selection criterion, number of tuning runs, and resulting configuration).

**R3.9 Comment:** *The paper lacks discussion on sim-to-real transfer challenges, such as domain shift and hardware constraints.*

**Response:** A sim-to-real / limitations discussion was added.

**Action taken:** New “Limitations and Future Work” subsection discussing the sim-to-real gap: domain shift from inaccurate mass/inertia identification, unmodeled sensor noise and latency, actuator constraints, and onboard compute limits, with directions for domain randomization and hardware-in-the-loop validation.

**R3.10 Comment:** *Some sections (especially reward equations and KPI definitions) are dense; adding intuitive explanations or diagrams would improve readability.*

**Response:** The dense sections were made more readable.

**Action taken:** The nine KPIs are now organized into three intuitive groups (tracking accuracy: K2/K3; base stability and safety: K4–K6; efficiency: K7–K9, with K1 as an overall scalar summary), each introduced in plain language before the formal definitions.

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*We believe the revised manuscript fully addresses the reviewers' concerns and thank them again for helping us improve the paper.*